ART FIBER SYSTEMS CHALLENGE · SCHUBKI NATIONAL AI HACKATHON 2026

# AFS Spectroscopy Assistant

*A customer enquiry goes in. A cited first technical assessment comes out.*

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## Team Helios

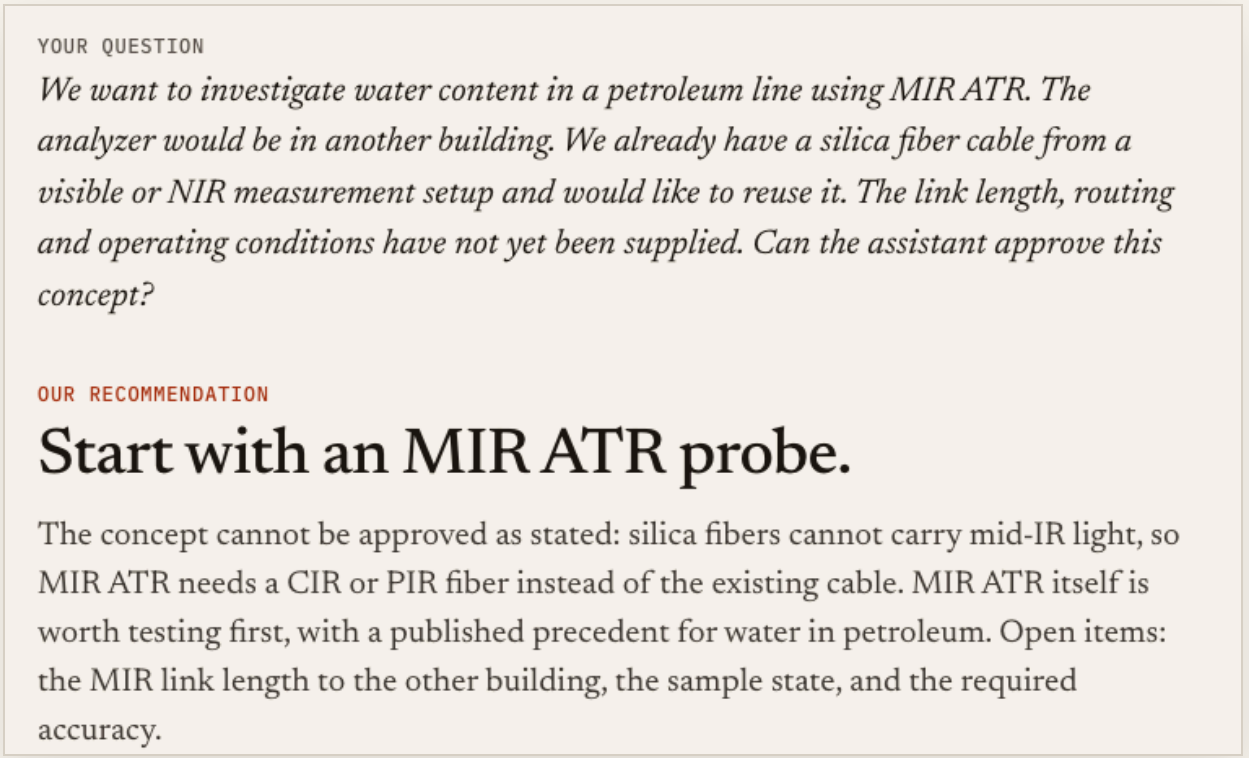
Rakibuzzaman Rahat, Md Mahmud Siddique


github.com/sleipnir029/Afs-Spectroscopy-Assistant

WHAT WE BUILT

# Ask in plain words. Get a cited answer.

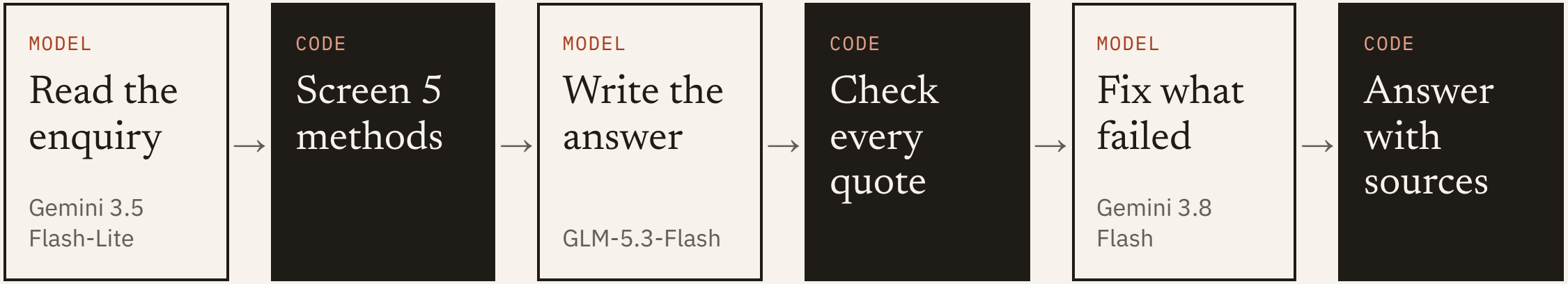
- Recommends the method and probe to test first.
- Every key claim cites the AFS page it comes from.
- Asks only the questions that would change the advice.
- Follow-ups in seconds; new facts update the answer.




Screenshot: the real app, evaluation case H04 (outputs/final/H04.json), unedited

SYSTEM ARCHITECTURE

# Models write. Code checks.




Three models, each on its own job. In a blind test of 7 setups, this mix was **3–6× faster** at close to the best quality, about **$0.03** per case.

USE OF AFS MATERIALS

# Built on the AFS documents

**67**

AFS pages the model reads

**44**

rules from the AFS documents, each tied to a quote

**120**/120

quotes found word for word on the cited page, in the six evaluation cases

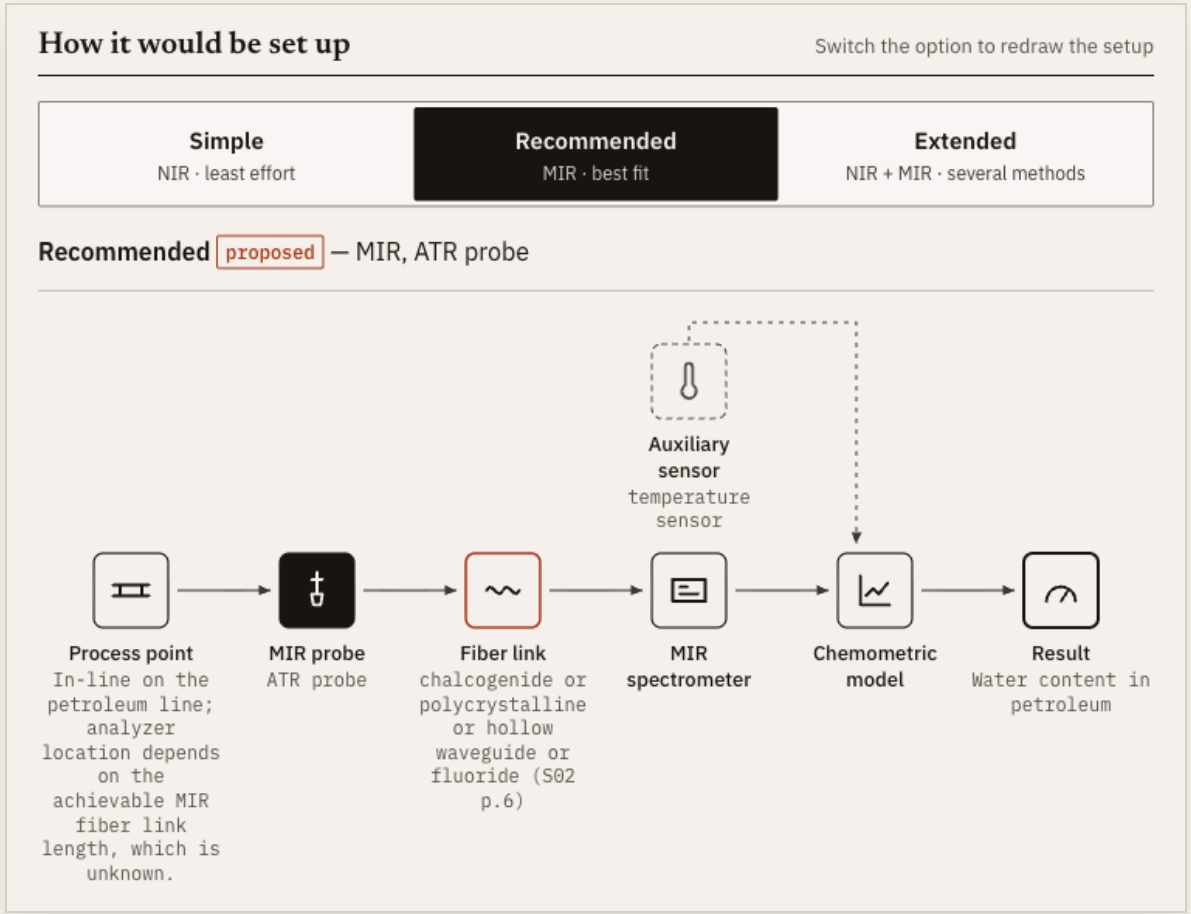
Code checks every quote against its page. Anything unsupported is removed or labelled.

EXAMPLE RECOMMENDATION · EVALUATION CASE H04, UNEDITED

# “Can we reuse our silica cable for mid-IR?”

THE AGENT’S ANSWER

Not as asked: silica fibers cannot carry mid-IR light. Test MIR ATR with a CIR or PIR fiber.




Screenshot: the real app, outputs/final/H04.json

EXAMPLE FEASIBILITY-STUDY PLAN · SAME CASE, UNEDITED

# It asks, then plans the first test

## QUESTIONS FOR THE CUSTOMER

### Help us sharpen this

3 answers would change the advice. Pick a question to fill in the customer's answer.

1 What is the actual link length and routing between the petroleum line and the analyzer building?

2 What is the physical state and condition of the petroleum at the measurement point (liquid, viscous, opaque, temperature, fouling)?

3 What water-content range, accuracy and response time does the investigation require?

Update the recommendation

Takes a few minutes; the new answer appears below with what changed.

## FIRST FEASIBILITY TEST

### First feasibility test

5 steps

1. **Collect representative petroleum samples spanning the expected water-content range and physical variation (temperature, opacity, phase).** Ensure the calibration space covers real line conditions.
2. **Measure each sample with a laboratory MIR ATR setup (ZrO2 or diamond ATR tip) using a CIR/PIR fiber probe.** Establish whether MIR ATR spectra distinguish the water levels of interest. S02 p.13
3. **Obtain independent reference water-content values for every sample by a laboratory method (e.g. Karl Fischer titration).** Provide the ground truth for calibration and validation.
4. **Build and validate a chemometric calibration on held-out samples; if MIR ATR alone falls short of the target, test the MIR+NIR combination against the best single method.** Quantify achievable accuracy and decide whether fusion adds value. S01 p.5
5. **Confirm the achievable MIR fiber link length with AFS before fixing the analyzer location; define go/no-go criteria (accuracy, detection limit, response time) with the customer.** The remote-analyzer concept depends on a link length no source specifies.

| | |
|---|---|
| Reference measurements | Karl Fischer titration or an equivalent laboratory water-in-oil method, supplied by the customer. |
| Decision basis | Go/no-go criteria (required accuracy, detection limit, response time) to be defined numerically by the customer; the plan proceeds only if the MIR link length is confirmed feasible. |

Screenshots: the real app, outputs/final/H04.json

RESULTS

# Six evaluation cases, answered by the agent

### Reuse an NIR calibration on a new blend?

**Not supported.** Rebuild the calibration.

### A clear and an opaque dairy stream?

**Provisional.** A different NIR probe for each.

### Fluorescence for glucose?

**Not supported.** Test Raman first.

### Silica cable for mid-IR?

**Not supported.** Use a CIR or PIR fiber.

### Unknown contaminant, sensor now?

**Not enough information.** Lab trial first.

### A kidney study as proof for surgery?

**Not supported.** Test on own tissue first.

**120/120** quotes verified  **1.5–3** min per case  **$0.29** for all six

outputs/final/H01–H06 (.md, .json, brief/) · run once, not edited

LIMITATIONS

# Limitations and next steps

| | |
|---|---|
| Full answers are long (~3,000 words) | → A short customer reply |
| A quote on the page is not yet proof | → A second checker for support |
| Confidence is mostly “Low” | → Confidence that tracks the facts |
| A run takes 1.5–3 minutes | → A faster fixing step |

Code, answers and demo video (3:27): github.com/sleipnir029/Afs-Spectroscopy-Assistant · Team Helios

outputs/final/H0*.md, H0*.json · README.md “Known limitations”